AP PHYSICS 1

Unit 1 • Worksheet 1.1A • Answer key

Worksheet 1.1A • Scalars and Vectors in One Dimension

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

Which of the following quantities is a vector?

(A) Speed
(B) Distance
(C) Displacement
(D) Time

Solution

Answer: (C). Displacement has a magnitude and a direction. Speed, distance and time have a magnitude only.

Question 2

Which of the following quantities is a scalar?

(A) Velocity
(B) Acceleration
(C) Force
(D) Mass

Solution

Answer: (D). Mass has no direction. Velocity, acceleration and force are vectors.

Question 3

An object moves along the x-axis from $x = +3\,\text{m}$ to $x = -5\,\text{m}$. What is its displacement?

(A) $+8\,\text{m}$
(B) $-8\,\text{m}$
(C) $-2\,\text{m}$
(D) $+2\,\text{m}$

Solution

Answer: (B). $\Delta x = x_f - x_i = -5 - (+3) = -8\,\text{m}$.

Question 4

A person walks 30 m east and then 50 m west. What are the distance traveled and the displacement?

(A) 80 m; 20 m west
(B) 80 m; 20 m east
(C) 20 m; 80 m west
(D) 20 m west; 80 m

Solution
Answer: (A). Distance $= 30 + 50 = 80\,\text{m}$. Displacement $= +30 - 50 = -20\,\text{m}$, that is 20 m west.

Question 5

East is positive. Which velocity has the greatest speed?

(A) $+6\,\text{m/s}$
(B) $-9\,\text{m/s}$
(C) $+8\,\text{m/s}$
(D) $-4\,\text{m/s}$

Solution
Answer: (B). Speed is the magnitude: $9\,\text{m/s}$ is the largest. The minus sign only means the motion is to the west.

Question 6

A ball is thrown straight up and caught at the same height. What is its displacement for the whole trip?

(A) Twice the maximum height
(B) Zero
(C) The maximum height
(D) Negative

Solution
Answer: (B). It ends where it started, so $\Delta y = 0$, even though the distance traveled is twice the maximum height.

Question 7

Up is positive. A stone falls 12 m from the top of a cliff to the ground. What is its displacement?

(A) $+12\,\text{m}$
(B) $0\,\text{m}$
(C) $-12\,\text{m}$
(D) $24\,\text{m}$

Solution
Answer: (C). It moves 12 m in the negative (downward) direction.

Question 8

Two forces act on a box along the same line: 40 N to the right and 25 N to the left. What is the net force?

(A) 65 N to the right
(B) 15 N to the left
(C) 15 N to the right
(D) 65 N to the left

Solution
Answer: (C). $+40 - 25 = +15$ N: to the right.

Question 9 (Free response)

A car on a straight road starts at $x = 0$, drives to $x = +120$ m, then back to $x = +40$ m, then forward to $x = +70$ m.

(a) Find the total distance traveled.
(b) Find the displacement.
(c) Can the magnitude of a displacement ever be larger than the distance traveled? Explain.

Solution
(a) $120 + 80 + 30 = 230$ m. (b) $\Delta x = 70 - 0 = +70$ m. (c) No. The displacement is the straight-line change in position; the path is at least that long. They are equal only if the object never turns around.